ANIRUDH D

Sr Python Developer | AWS Certified Developer Associate | Email: desirajuanirudh.py@gmail.com | Contact: (916) 827-7411

SUMMARY

Sr. Python Developer with 6+ years of experience building web API s, data/ML pipelines, and automation across cloud-native environments. Strengths include Python (Flask/Fast-API), back-end architecture, AWS, CI/CD, and data engineering with Pandas/NumPy/scikit-learn/Spark, complemented by hands-on UI work (React/HTML/CSS/JS). Proven track record integrating applications with Oracle/MySQL/PostgreSQL/MongoDB, optimizing performance, and enforcing secure, auditable workflows in enterprise and healthcare settings. Experienced with Agile/Scrum, test automation, and developer productivity tooling that lifts throughput and quality.

KEY ACHIEVEMENTS

- Delivered production-grade Flask services and REST API s powering data ingestion, transformations, and reporting, standardized API contracts and elevated reliability under variable load.
- Automated acquisition-to-testing workflows with Python, reducing manual effort via validation, schema enforcement, and traceability artifact generation, integrated with source control and CI/CD for repeatable releases.
- Built data/ML utilities (Pandas, scikit-learn/OpenCV) and NLP pipelines for classification, similarity scoring, and analytics, exposed services through Flask endpoints.
- Modernized delivery with Git/Jenkins/Azure DevOps/GitHub Actions, containerization, and infrastructure automation on AWS (Lambda, API Gateway, EC2, S3, RDS).

TECHNICAL SKILLS

- **Frameworks / Backend:** Flask, REST APIs, MVC, experience integrating React UIs with Flask/FastAPI backends
- **Web Technologies:** HTML5, CSS3, JavaScript, jQuery, ReactJS, XML, NodeJS
- **Programming Languages:** Python, SQL/PL-SQL, Java, C++
- **Scripting:** Bash/Unix Shell, PowerShell
- **Cloud / Infrastructure:** AWS (EC2, S3, RDS, Lambda, API Gateway, ELB), Azure (exposure)
- **Data & Machine Learning:** Pandas, NumPy, scikit-learn, OpenCV, Spark, NLP (NLTK, TF-IDF, tokenization, stemming, similarity)
- **Databases:** MySQL, PostgreSQL, MongoDB, exposure to Teradata connectors, HDFS ingestion
- **DevOps / CI-CD:** Git (GitHub, Bitbucket, Subversion), Jenkins, Maven, Docker, Kubernetes (experience), Bamboo (exposure)
- **Testing / QA: U**nittest, Selenium WebDriver, JIRA , compatibility and cross-browser testing, JAMA, ALM, DOORS
- **Methodologies:** Agile, Scrum, Waterfall, Test-Driven Development (TDD) in select projects

PROFESSIONAL EXPERIENCE

Client: GE Healthcare, Waukesha, WI **Sep'25 – Till Date**

Role: Python Developer

- Designed and developed Python-based engineering automation workflows spanning CT data acquisition, integration, validation, and testing, improving workflow repeatability and reducing manual execution across engineering processes.
- Developed Python and Pandas ETL/data-processing pipelines to ingest, transform, enrich, and validate CT acquisition datasets, implementing schema enforcement and automated quality checks for reliable downstream testing and analysis.
- Automated CT system validation and regression testing using Python, including phantom-based data acquisition, calibration verification, performance validation, and repeatable test execution across engineering workflows.
- Integrated Python services with requirements and test-management platforms, including ALM, JIRA, and JAMA-style systems, to automate requirements traceability, test-status reporting, and audit-ready requirement-to-test matrices for Systems and V&V stakeholders.
- Developed reusable Python engineering utilities for CT image analysis, reconstruction optimization, dataset management, and test-data processing, supporting engineers and imaging scientists throughout validation and development workflows.
- Supported full-stack integration and internal engineering dashboards using React, JavaScript, HTML5, CSS3, and Bootstrap, enabling visualization of test results, workflow status, acquisition data, and operational metrics.
- Collaborated with CT Engineering, Systems Engineering, V&V, imaging scientists, and product stakeholders to translate requirements into scalable software architectures, document technical designs and data flows, and support validation in a regulated medical-device engineering environment.

Environment: Python, Flask, AWS (EC2, S3, RDS, Redshift, Lambda, API Gateway, ELB), Oracle ATP, MongoDB, MySQL, ReactJS, HTML5/CSS/JS, Selenium, Pandas, JIRA, Linux, C++.

Client: Intel, Folsom, CA **Nov'21 – Aug'25**

Role: Python Developer

- Supported pre-silicon GPU validation and hardware modeling for display interfaces (DP/eDP/HDMI), scripting Python-based tooling to accelerate verification flows and data analysis.
- Designed and maintained Python applications/automation with an emphasis on scalability, maintainability, and performance across environments led Python 3.x upgrades and dependency compatibility fixes.
- Built NLP pipelines (NLTK, scikit-learn, TF-IDF, cosine similarity) for text processing, classification, and clustering, exposed analytics via Flask REST endpoints for downstream consumers.
- Engineered advanced text-processing and vectorization for structured/unstructured datasets implemented robust preprocessing, tokenization, and feature extraction stages.
- Deployed Python/NLP services in enterprise environments with attention to security, performance tuning, and configuration management integrated CI/CD and artifact management.
- Contributed to full-stack features (Python, Java, ReactJS, HTML/CSS/JS) and dashboard components practiced Agile delivery and used JIRA for backlog/defect tracking.
- Implemented compatibility and UI tests using Selenium/WebDriver and cross-browser validation, capturing dynamic/static DOM behavior using HTML IDs/XPath selectors.

Environment: Python, Flask, AWS, MongoDB, Pandas, Selenium, ReactJS, Java, HTML5/CSS/JS, JIRA, Linux MySQL.

Client: Savitude, Palo Alto, CA **Jan'21 – Oct'21**

Role: Python Developer — Project: Fashion Dashboard

- Trained and served ML models to predict garment pixel positions on a mannequin figure operationalized inference on AWS Lambda and returned result images per user input.
- Implemented service-oriented architecture with Python Flask backends and React/HTML/CSS frontends, enforced API versioning and validation for reliable UI integration.
- Built Selenium regression suites integrated WebDriver tests with the API test framework to catch UI/contract drift early.
- Employed Pandas for time-series/tabular processing and third-party data cleanup practiced TDD to harden critical code paths.
- Developed cloud data flows on AWS EC2/S3/RDS (PostgreSQL/MySQL) used Flask ORM for data access and wrote Oracle PL/SQL procedures and triggers for reporting logic.
- Managed dashboards for customers/admins (Flask, Oracle DB, PostgreSQL, VMware APIs) automated builds and promotions with Python/Groovy, Jenkins, and Tomcat deploys.

Environment: Python, Flask, Selenium, HTML5/CSS, AWS (EC2, S3, RDS, Lambda), Oracle DB, PostgreSQL, Pandas, Spring, MS SQL Server, Jasper Reports, JavaScript, Eclipse, Linux, Shell, REST/MVC.

Client: Stamp It Biz, Hyderabad, India **Jan'18 – Aug'18**

Role: Full Stack Developer Intern — Project: Internal Website

- Participated across SDLC phases using Agile implemented MVC with Flask and RESTful APIs for internal services.
- Owned database integration and tuning applied ORM patterns for Oracle schemas and optimized stored procedures/queries for performance.
- Automated server-side tasks in Node.js connecting SQL/NoSQL datastores, configured infrastructure with SaltStack for predictable environments.
- Established CI with Jenkins and source control with GitHub implemented automated Selenium tests and unit tests (unittest) to reduce regressions.
- Deployed SOAP web services on Tomcat documented issues and progress in JIRA and resolved defects using browser dev tools.

Environment: Python, Flask, HTML/CSS/JavaScript, NoSQL, jQuery, Sublime Text, JIRA, Git, pyBuilder, unittest, Web Services.

PROJECTS

Acquisition to Test Orchestrator (Pipelines + Testing) | AI-Driven Test Failure Root Cause Analyzer (Enterprise + GenAI) | Keyboard Logger

EDUCATION

Master of Computer Science, California State University, Sacramento, California.